

House Price Prediction – Project Summary

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics and to identify the factors that have the greatest influence on house prices.

Dataset

The Housing Prices Dataset was obtained from Kaggle and contains 545 records with 13 attributes, including property size, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other housing-related features.

Data Preprocessing

The following preprocessing steps were performed:

- Loaded and explored the dataset using Pandas.
- Checked for missing values and confirmed data quality.
- Checked for duplicate records and removed them if present.
- Converted categorical variables into numerical format using one-hot encoding.
- Prepared the data for machine learning model training.

Models Used

Two regression models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

Model Performance

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,015,683	1,393,448	0.616

Based on the evaluation metrics, Linear Regression achieved better performance than Random Forest Regressor on this dataset.

Key Findings

Feature importance analysis revealed that the following factors had the strongest impact on house prices:

- Number of Bathrooms
- Air Conditioning Availability
- Hot Water Heating
- Preferred Area
- Number of Stories
- Basement Availability
- Main Road Access
- Parking Availability

The correlation analysis also indicated that larger and better-equipped houses generally tend to have higher market values.

Visualizations Created

The project included the following visualizations:

1. Distribution of House Prices (Histogram)
2. Correlation Heatmap
3. Actual vs Predicted House Prices Scatter Plot

These visualizations helped in understanding data distribution, relationships between features, and model prediction performance.

Conclusion

A machine learning-based house price prediction system was successfully developed using the Housing Prices Dataset. After preprocessing the data and training multiple regression models, Linear Regression emerged as the best-performing model with an R^2 score of approximately 0.653. The analysis demonstrated that property amenities and structural characteristics significantly influence house prices. The developed model can assist buyers, sellers, and real estate businesses in making more informed property valuation decisions.